Preferences for Product Variety in Targeted Pricing

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Roadmap

1. Motivation and contribution
2. Data and descriptive evidence
3. Monte Carlo motivation
4. Demand estimation and results
5. Pricing and learning
6. Conclusions and discussion

Roadmap: Motivation and contribution

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Can firms price the desire for a change?

- Repeated consumption may reduce the appeal of a flavor.
- Purchase histories can reveal when a consumer is ready to switch.
- A retailer can use that information to optimally discount.

Research question

How do preferences for variety shape demand and the value of consumer information?

Two exercises, two takeaways

- **Demand:** flavor preferences change little when fatigue is included; some fatigue parameters are weakly identified.
- **Pricing simulation:** knowing switching types raises per-unit profits by up to about 2 cents in the reported normalization.

Contribution: variety across attributes

- Study flavor switching within brands as well as across brands.
- Separate baseline flavor preferences from dependence on purchase history.
- Connect switching preferences and state dependence to the information a retailer uses for pricing.

Roadmap: Data and descriptive evidence

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Why yogurt, and whose purchases?

- Nielsen HomeScan and MarketScan, 2022–2024.
- Eight markets: Atlanta, Chicago, Denver, Houston, Des Moines, Phoenix, Philadelphia, San Diego.
- Single-person households reduce ambiguity about whose tastes drive purchases.
- Single-serving yogurt offers flavor variation and limited storability.

The estimation sample contains 998 households

	Full sample	Estimation sample
Households	5,408	998
Mean store trips / household	308.15	400.05
Mean observed yogurt choice weeks	—	3.91
Coupon / deal purchase rate	26.52%	23.10%
Mean weekly yogurt units (purchasers)	—	8.44

Implication

The panel contains limited yogurt history for many households.

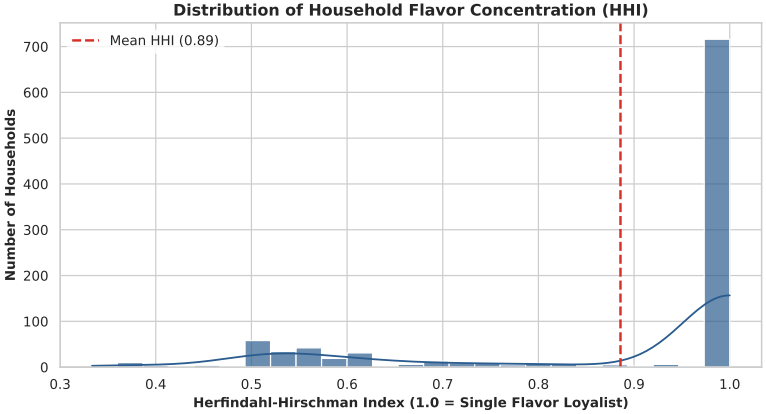
Three flavor groups simplify a rich product space

- **Berry:** berry-labeled flavors plus cherry.
- **Plain:** a separate category.
- **Other:** all remaining flavors.
- **Outside option:** no yogurt purchase.

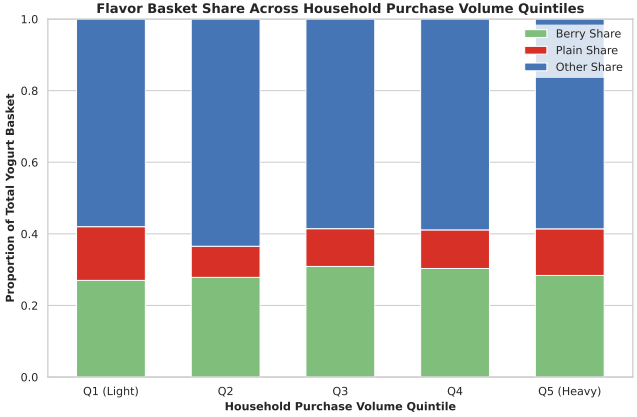
Tradeoff

Aggregation makes estimation tractable but hides switching within a flavor group.

Many households concentrate on one flavor group



Other flavors dominate across purchase volumes



Roadmap: Monte Carlo motivation

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Monte Carlo: simulation procedure

Simulation design	Assigned value
Replications / evaluation periods / burn-in	1,000 / 100 / 5
Product attributes / baseline taste	$X_j \in \{1, 2, 3, 4, 5\}$; $\beta = 2$
Variety grid / choice shocks	$\gamma \in \{0, 3, 6, 9\}$; $\varepsilon_{jt} \sim \mathrm{T1EV}$

1. For each γ, initialize consumption histories during burn-in.
2. At each evaluation date, combine baseline attribute utility with the history-dependent distance reward and a choice shock.
3. Record the selected product, update its consumption history, and repeat.
4. Summarize choice probabilities across the simulated runs.

Purpose: show how variety changes choices in a controlled setting.

Monte Carlo: simulated choice probabilities

	$\gamma = 0$	$\gamma = 3$	$\gamma = 6$	$\gamma = 9$
Product 1 ($X = 1$)	.0016	.1352	.3321	.3973
Product 2 ($X = 2$)	.0084	.1325	.0541	.0084
Product 3 ($X = 3$)	.0437	.0844	.0026	.0000
Product 4 ($X = 4$)	.1934	.0553	.0023	.0002
Product 5 ($X = 5$)	.7529	.5926	.6089	.5941

Interpretation

Without variety, the highest-attribute product dominates. With stronger variety, choices concentrate at the endpoints.

This motivates modeling history; it does not establish identification in the yogurt panel.

Roadmap: Demand estimation and results

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Demand combines stable tastes and purchase history

$$U_{ijt} = \underbrace{\beta_0 + \beta_i' X_{jt}}_{\text{baseline tastes}} + \underbrace{\gamma_i \Xi_{ijt}}_{\text{history dependence}} + \underbrace{\alpha p_{jt}}_{\text{price}} + \varepsilon_{ijt}.$$

- Inside choices: berry, plain, other; outside utility normalized to zero.
- Ξ_{ijt} summarizes prior consumption and switching.
- Negative fatigue coefficients reduce utility as the history index rises.

Defining the consumption-history index Ξ_{ijt}

$$\Xi_{ijt} = \underbrace{\lambda \sum_{s=1}^{t-1} \delta^{t-s} \Xi_{ijs}}_{\text{discounted past index values}} + \underbrace{(1-\lambda) \sum_{k \in \{B,P,O\},\, k \neq j} \rho_k x_{ik,t-1}}_{\text{previous-period purchases of other flavors}}$$

- i: household; j: candidate flavor; t: observed choice period; $s < t$: past periods.
- $x_{ik,t-1} = q_{ik,t-1} / \sum_{h \in \{B,P,O\}} q_{ih,t-1}$: flavor k's share of prior yogurt units; set all shares to zero if no yogurt is purchased.
- $\rho_k = 1$: equal cross-flavor weights; outside purchases do not enter the sum.
- $\lambda = 0.7$: memory weight; $\delta = 0.9$: discount per observed period.
- Initialize $\Xi_{ij1} = 0$ and prior shares at zero; compute the index before period-t choices.

How Ξ_{ijt} changes: a two-period example

Start with zero histories. Suppose the first observed basket contains only berry:

$$(x_{iB1}, x_{iP1}, x_{iO1}) = (1, 0, 0), \qquad (\Xi_{iB2}, \Xi_{iP2}, \Xi_{iO2}) = (0, 0.3, 0.3).$$

If the second basket contains only plain, then at the next choice:

$$\begin{aligned}
\Xi_{iB3} &= 0.7(0.9 \times 0) + 0.3 = 0.3, \\
\Xi_{iP3} &= 0.7(0.9 \times 0.3) + 0 = 0.189, \\
\Xi_{iO3} &= 0.7(0.9 \times 0.3) + 0.3 = 0.489.
\end{aligned}$$

Interpretation of this specification

A negative γ_j lowers candidate-flavor utility when its index rises. Because other-flavor purchases raise the index, this is not a pure own-flavor fatigue stock.

What separates tastes from fatigue?

- **Stable tastes:** persistent differences in flavor choices.
- **Fatigue:** changes in choices associated with a household's own history.
- **Challenge:** heterogeneity, prices, and unobserved shocks can also generate switching.
- **Limited history and coarse groups** make the distinction difficult.

Demand estimation: first-stage price control function

$$p_{jmt} = a + b \underbrace{\bar{p}_{j,-m,t}}_{\text{other-market price}} + \tau_t + \eta_{jmt}, \qquad \hat{\eta}_{jmt} = p_{jmt} - \hat{p}_{jmt}.$$

1. Match household choices to flavor-specific prices and purchase histories.
2. Regress local prices on other-market prices and time controls.
3. Include the fitted residual in utility to account for endogenous prices.

Required exclusion

Other-market prices must shift local prices without sharing unobserved local demand shocks, conditional on the controls.

Demand estimation: complete utility specification

For $j \in \{B, P, O\}$ (berry, plain, other), let $D_j^f = \mathbf{1}\{j = f\}$:

$$
\begin{aligned}
U_{ijt} &= V_{ijt} + \varepsilon_{ijt}, \\
V_{ijt} &= \beta_0 + \beta_i^B D_j^B + \beta_i^P D_j^P \\
&\quad + \gamma_i^B D_j^B \Xi_{ijt} + \gamma_i^P D_j^P \Xi_{ijt} + \alpha p_{j,m(i),t} + \sigma \hat{\eta}_{j,m(i),t}.
\end{aligned}
$$

- Other flavor is the reference; outside deterministic utility is zero.
- Ξ_{ijt} summarizes own past consumption and alternative-flavor history.
- History weight $\lambda = 0.7$ and decay $\delta = 0.9$ are fixed.
- Random coefficients allow persistent household differences in tastes and fatigue.

Demand estimation: panel likelihood and mixed logit

Conditional on household coefficients $b_i = (\beta_i^B, \beta_i^P, \gamma_i^B, \gamma_i^P)$:

$$P_{ijt}(b_i) = \frac{e^{V_{ijt}(b_i)}}{1 + \sum_{k \in \{B,P,O\}} e^{V_{ikt}(b_i)}}, \qquad P_{i0t}(b_i) = \frac{1}{1 + \sum_{k} e^{V_{ikt}(b_i)}}.$$

$$L_i(b_i) = \prod_{t \in \mathcal{T}_i} \prod_{j \in \{0,B,P,O\}} P_{ijt}(b_i)^{y_{ijt}}, \qquad y_{ijt} = \mathbf{1}\{\text{observed choice is } j\}.$$

$$\boxed{\hat{\theta} = \arg\max_{\theta} \sum_i \log\left[\int L_i(b)\, dF(b;\theta)\right]}$$

Simulation approximation: $\int L_i(b) dF(b;\theta) \approx R^{-1} \sum_{r=1}^{R} L_i(b_i^{(r)})$.

Standard logit uses fixed coefficients: $\ell(\theta) = \sum_{i,t,j} y_{ijt} \log P_{ijt}(\theta)$.

Demand estimation: four specifications

	No fatigue	With fatigue
Standard logit	Fixed β; $\gamma = 0$	Fixed β, γ
Random coefficient logit	Heterogeneous β; $\gamma = 0$	Heterogeneous β, γ

- Price and the control-function residual enter all four specifications.
- Compare baseline flavor tastes and price effects across columns.
- Assess fatigue means and dispersion separately from optimizer convergence.
- Identification depends on within-household history variation, valid price instruments, and the treatment of initial histories.

Baseline demand estimates barely move

	Standard logit		Random coefficients	
	No fatigue	Fatigue	No fatigue	Fatigue
Berry preference	−.7635	−.7576	−.7632	−.7489
Plain preference	−1.3899	−1.3897	−1.3930	−1.3956
Price	−.9836	−.9821	−.9836	−.9804
Control function	.1670	.1667	.1654	.1662

Main empirical result

The largest baseline flavor change is about 1.9% for berry in the random coefficient model.

Berry fatigue is negative; plain is imprecise

	Standard logit	Random coefficients
Berry fatigue	−.7262	−7.2030
Standard error	(.1485)	(1.5424)
Plain fatigue	−.0351	−.0555
Standard error	(.1009)	(.1032)

- Berry has a negative estimated history coefficient.
- Plain fatigue is not statistically distinguishable from zero.
- Large changes across specifications warrant caution.

Heterogeneity exposes an estimation problem

Random coefficient SD	Estimate	SE
β_{berry}	1.0513	.0916
β_{plain}	1.0513	.1492
γ_{berry}	148.4132	155.3868
γ_{plain}	1.0513	.1641

Diagnostic

The berry fatigue dispersion is extremely large and imprecisely estimated.

Optimization success alone does not establish reliable identification.

Dollar interpretations inherit parameter uncertainty

Magnitude in dollars	Standard	Random coefficients
Berry preference	.77	.76
Plain preference	1.42	1.42
Berry fatigue	.74	7.35
Plain fatigue	.04	.06

$$\text{Dollar magnitude per index unit} = \frac{|\text{utility coefficient}|}{|\alpha|}.$$

Interpretation

The large \$7.35 number belongs to berry fatigue and is sensitive to the specification.

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From empirical demand to a pricing mechanism

- The demand exercise suggests a role for history, with substantial identification limits.
- The supply exercise asks how a retailer would price if switching types were known or inferred.
- Two products, two types, two periods.
- Parameters are assigned; the exercise is not estimated from the demand results.

A retailer prices two switching types

$$V_{rjt} = \beta_{0j} + \gamma_r \mathbf{1}\{j \neq j_{t-1}\} + \alpha p_{rjt}, \qquad r \in \{H, L\}.$$

$$p_{rjt} = \bar{p}_j - d_{rjt}, \qquad m_{rjt} = p_{rjt} - c_j.$$

- High types value switching more: $\gamma_H > \gamma_L$.
- Logit probabilities include an outside option.
- A discount raises demand but reduces the margin per sale.

Prices affect both current and future profits

$$W_r(j) = \max_{\mathbf{p}_2(j)} \sum_k s_{rk2}(j)\big(p_{rk2}(j) - c_k\big),$$

$$\Pi^{\text{full}} = \sum_r N_r \sum_j s_{rj1}\left[p_{rj1} - c_j + \delta W_r(j)\right].$$

Perfect information

The retailer conditions prices on type and purchase history.

A lower price today can move a consumer into a more profitable state tomorrow.

Without type information, purchases update beliefs

$$q = \Pr(r = H), \qquad \tilde{s}_{j1} = q s_{Hj1} + (1 - q) s_{Lj1},$$

$$q_2(j) = \Pr(H \mid j) = \frac{q s_{Hj1}}{q s_{Hj1} + (1 - q) s_{Lj1}}.$$

- Period-one prices are common across unobserved types.
- Period-two prices can depend on observed purchase history.
- A choice relatively more likely for high types raises the posterior.

The information constraint changes the pricing problem

$$\Pi^{\text{limited}} = N \sum_j \tilde{s}_{j1}(\mathbf{p}_1) \left[p_{j1} - c_j + \delta W\big(j, q_2(j; \mathbf{p}_1)\big) \right].$$

1. Current margin versus current demand.
2. How purchases change future consumption states.
3. How purchases reveal switching type.

Mechanism

Prices influence both allocation and the informativeness of observed choices.

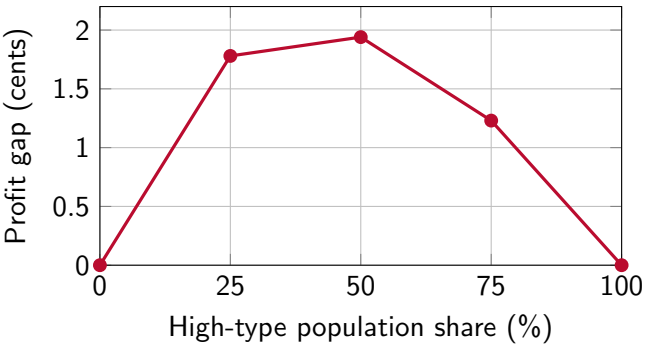
Simulation parameters are illustrative

Baseline product utilities	$(2.2, 2.0)$
Switching rewards (γ_H, γ_L)	$(2.0, 0.1)$
Price coefficient α	-1.5
Discount factor δ	0.95
Unit costs (c_1, c_2)	$(1.2, 1.0)$
Initial product-state weights	$(0.5, 0.5)$
Price bounds	$[\$1, \$8]$

Comparison

Vary the high-type population share under full and limited information.

The profit gap peaks in the mixed population



Reported value of information

Full-information profit minus limited-information profit.

Largest gap: 1.94 cents at a 50/50 type mix.

Pricing exercise: full results

Information Structure, Optimal Pricing, and Value of Information Across Population Mixes

Population Mix	Imperfect Price (P_1)		Posterior Beliefs		Imperfect	Perfect	Value of
	Good 1	Good 2	G1	G2	Profit	Profit	Info
100% LOW	2.0633	1.8646	0.0000	0.0000	0.6031	0.6031	-0.0000
25% HIGH, 75% LOW	2.1847	1.9944	0.3727	0.3641	0.7820	0.7999	0.0178
50% HIGH, 50% LOW	2.2819	2.1049	0.6510	0.6448	0.9773	0.9966	0.0194
75% HIGH, 25% LOW	2.3542	2.1887	0.8528	0.8504	1.1811	1.1934	0.0123
100% HIGH	2.4067	2.2494	1.0000	1.0000	1.3902	1.3902	0.0000

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What does this say about privacy policy?

- Restricting type information can reduce retailer profits in this model.
- The loss is greatest when the population contains both switching types.
- Consumer surplus and total welfare are not reported.
- The exercise does not establish the effect of a specific privacy law or whether firms stop collecting data.

Conclusions and next steps

- **Demand:** baseline taste estimates are stable; fatigue heterogeneity remains difficult to identify.
- **Pricing:** purchase histories can help retailers target switching incentives.
- **Next:** refine flavor groups and histories, audit heterogeneity, strengthen price instruments, and calibrate supply to the demand data.

Discussion prompts

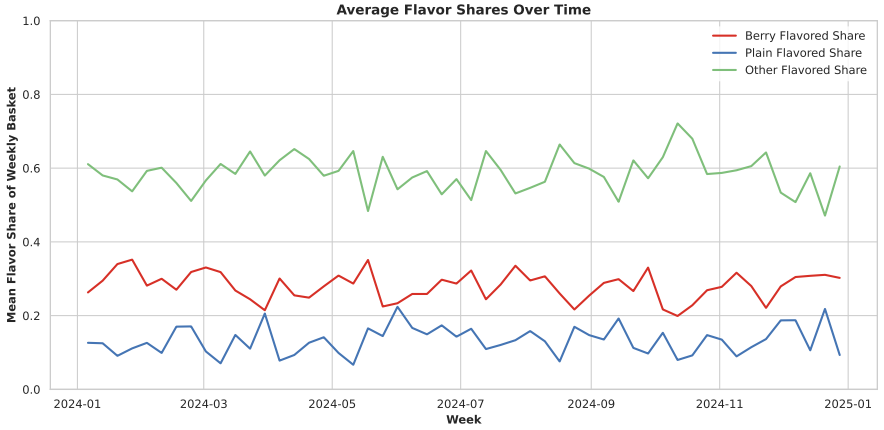
Questions for discussion

How can fatigue be distinguished from persistent unobserved tastes?

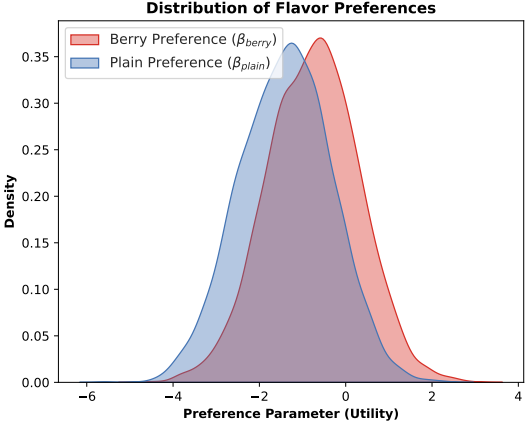
Which purchase histories are most informative about switching types?

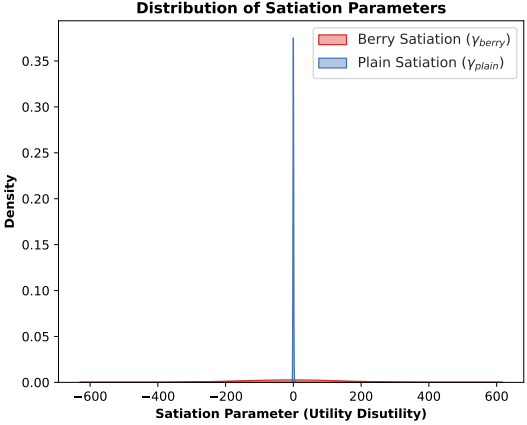
What consumer-welfare evidence is needed to evaluate privacy restrictions?

Backup: flavor shares over time

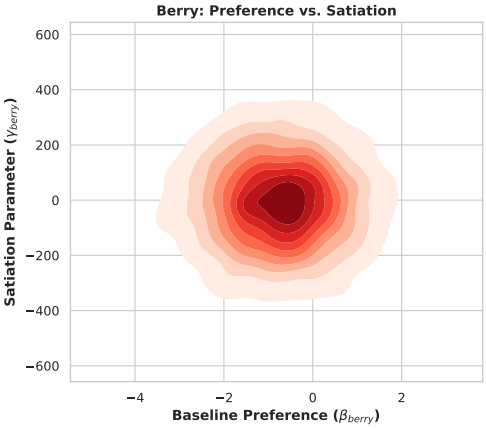


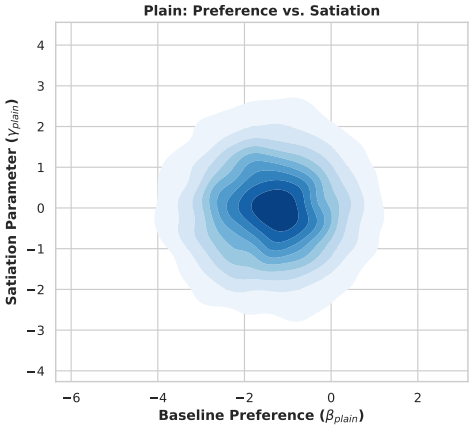
Backup: estimated marginal distributions





Backup: estimated joint distributions





Backup: selected literature

- Attribute satiation: McAlister; Lattin (1987).
- Variety and brand choice: Givon (1984); Chintagunta (1999).
- Bundles and variety: Dubé (2004).
- Price control functions: Petrin and Train (2010).
- Personalized pricing: Dubé and Misra (2023).

Full bibliographic entries appear in the accompanying paper.